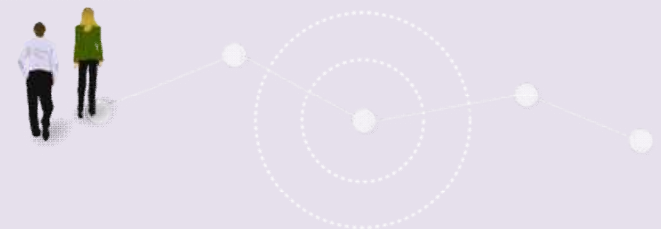


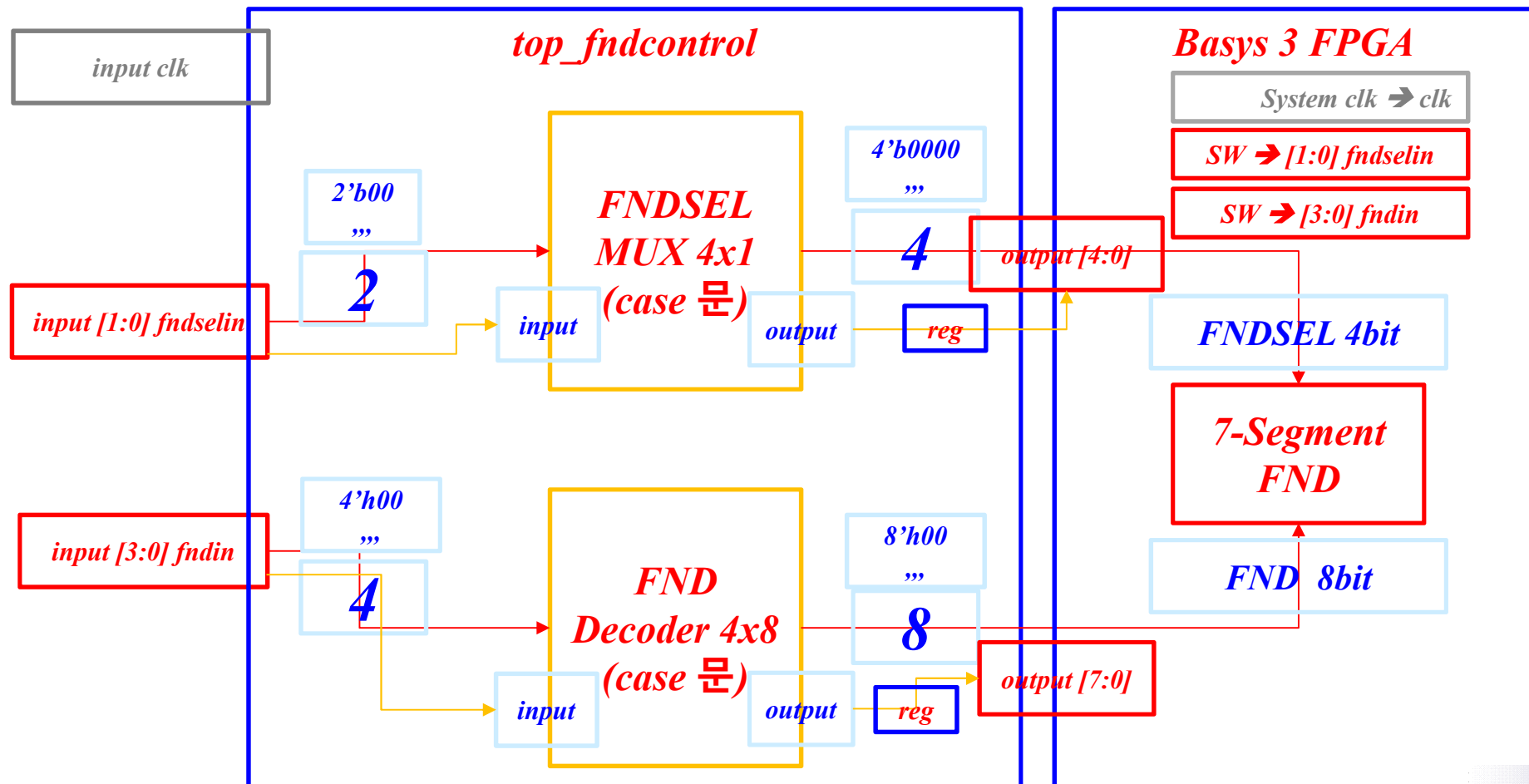
Example 01. *FND Controller I*



Example 01. *FND Controller I*

Example 01. FND Controller I

- **MUX**와 **Decoder**를 이용하여 **FND**를 숫자 9(or 16)를 4자리에서 각각 Display 할 수 있게 **FNDSEL**을 제어하여 구현



Example 01. FND Controller I

➤ *FND Selection* 구현을 위하여 4x1 Mux 를 이용하여 설계

1. `timescale 1ns / 1ps

2. module mux4x1(

3. input [1:0] s,

4. output reg [3:0] y

5.);

6. always @(s)

7. Begin : MUX4x1

8. case (s)

9. 2'b00: y = 4'b1110;

10. 2'b01: y = 4'b1101;

11. 2'b10: y = 4'b1011;

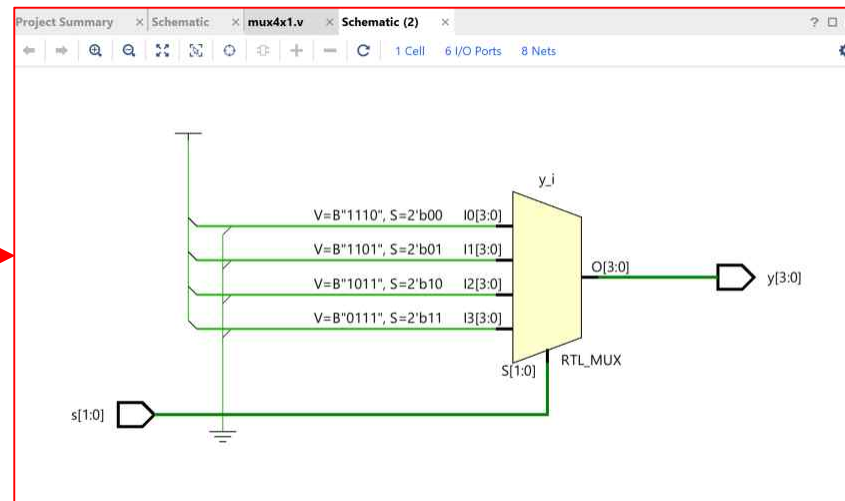
12. 2'b11: y = 4'b0111;

13. default: y = 4'b1111;

14. endcase

15. end

16. endmodule



1. `timescale 1ns / 1ps

2. module mux4x1(

3. input [1:0] x,

4. output reg [3:0] y

5.);

6. always @(x) begin

7. if (x == 2'b00)

8. y = 4'b1110;

9. else if (x == 2'b01)

10. y = 4'b1101;

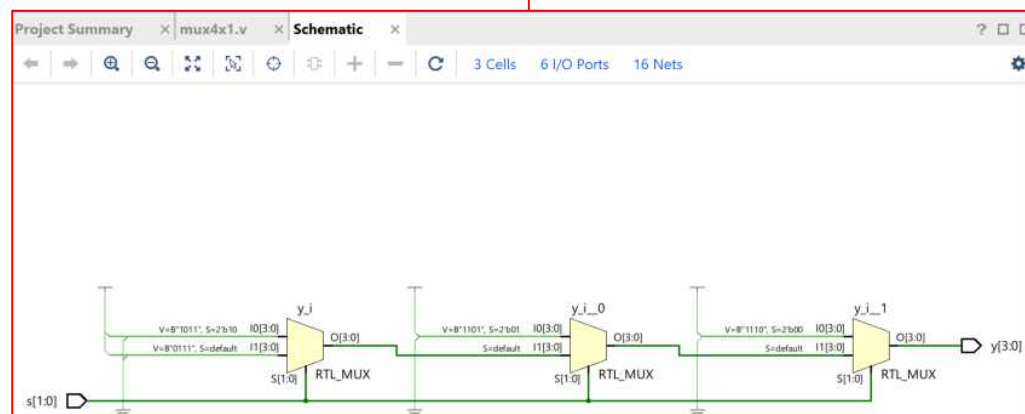
11. else if (x == 2'b10)

12. y = 4'b1011;

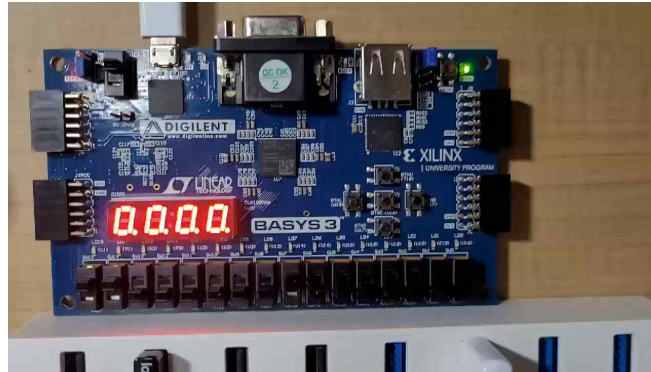
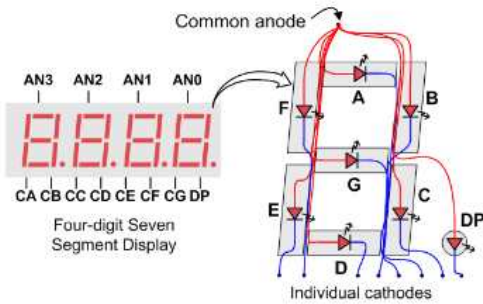
13. else y = 4'b0111;

14. end

15. endmodule



Example 01. FND Controller I



표시 값						출력 값									
2진수				10진수	16진수	dot	g	f	e	d	c	b	a	16진수	
0	0	0	0	0	0	0	1	0	0	0	0	0	0	40	
0	0	0	1	1	1	0	1	1	1	1	0	0	1	79	
0	0	1	0	2	2	0	0	1	0	0	1	0	0	24	
0	0	1	1	3	3	0	0	1	1	0	0	0	0	30	
0	1	0	0	4	4	0	0	0	1	1	0	0	1	19	
0	1	0	1	5	5	0	0	0	1	0	0	1	0	12	
0	1	1	0	6	6	0	0	0	0	0	0	1	0	02	
0	1	1	1	7	7	0	1	0	1	1	0	0	0	58	
1	0	0	0	8	8	0	0	0	0	0	0	0	0	00	
1	0	0	1	9	9	0	0	0	1	0	0	0	0	10	
1	0	1	0	10	a	0	0	1	0	0	0	0	0	20	
1	0	1	1	11	b	0	0	0	0	0	0	1	1	03	
1	1	0	0	12	c	0	0	1	0	0	1	1	1	27	
1	1	0	1	13	d	0	0	1	0	0	0	0	1	21	
1	1	1	0	14	e	0	0	0	0	0	1	0	0	04	
Kim Sung Woo				15	대한상공회의소 이력개발원	0	0	0	0	1	1	1	0	0e	

```

1. `timescale 1ns / 1ps

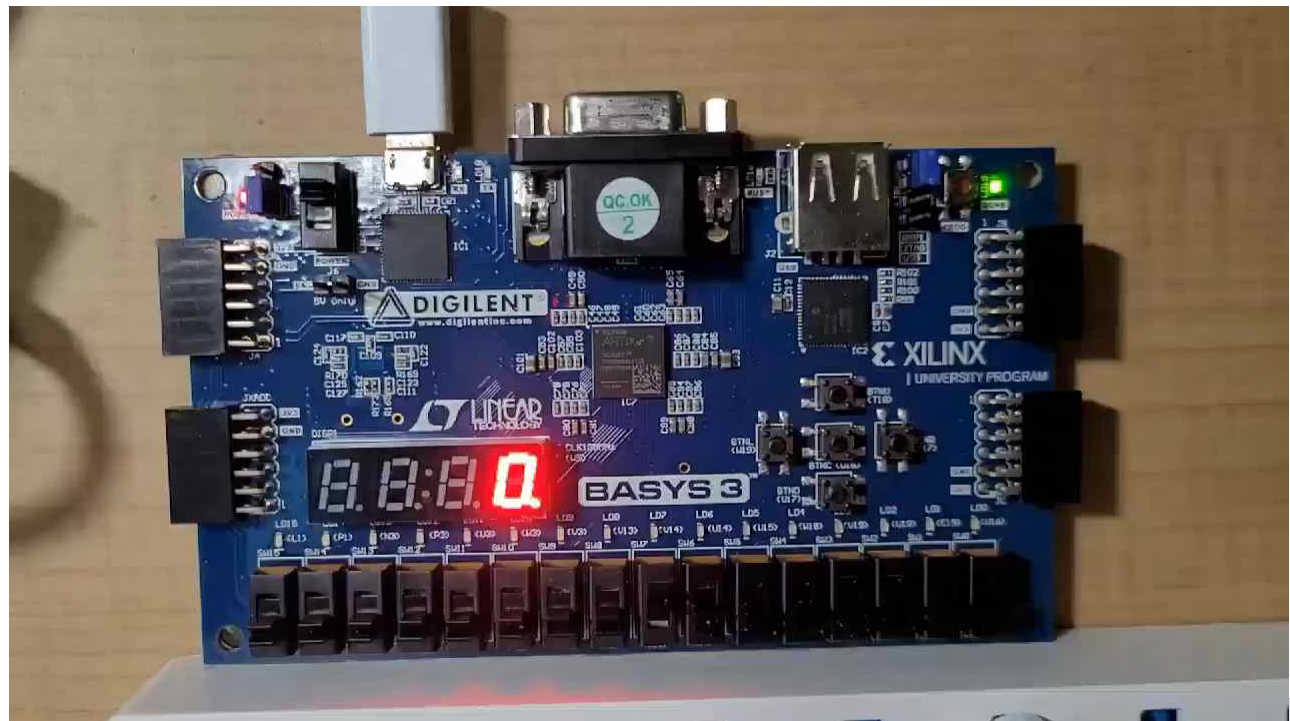
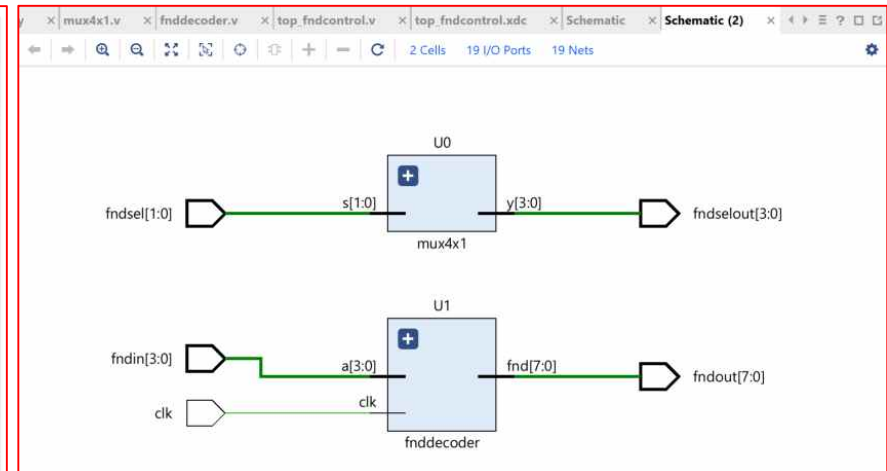
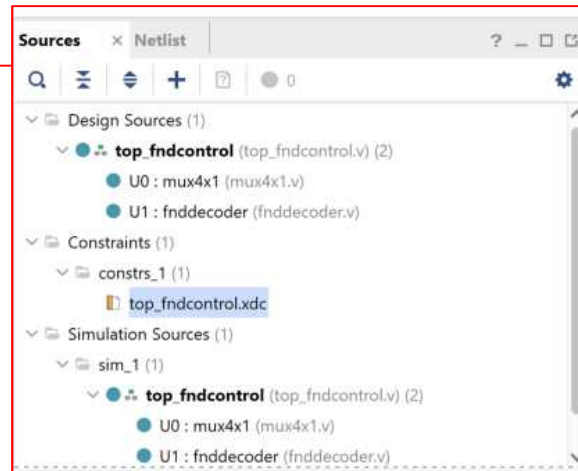
2. module fnddecoder(
3.     input [3:0] a,
4.     output reg [7:0] fnd );
5.     always @ (a) begin
6.         case (a)
7.             4'h0 : fnd = 8'h40;
8.             4'h1 : fnd = 8'h79;
9.             4'h2 : fnd = 8'h24;
10.            4'h3 : fnd = 8'h30;
11.            4'h4 : fnd = 8'h19;
12.            4'h5 : fnd = 8'h12;
13.            4'h6 : fnd = 8'h02;
14.            4'h7 : fnd = 8'h58;
15.            4'h8 : fnd = 8'h00;
16.            4'h9 : fnd = 8'h10;
17.            4'ha : fnd = 8'h20;
18.            4'hb : fnd = 8'h03;
19.            4'hc : fnd = 8'h27;
20.            4'hd : fnd = 8'h21;
21.            4'he : fnd = 8'h04;
22.            4'hf : fnd = 8'h0e;
23.            default: fnd = 8'hff;
24.        endcase
25.    end
endmodule

```

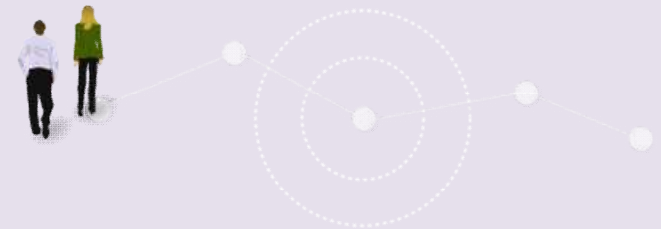
Example 01. FND Controller I

```

1. `timescale 1ns / 1ps
2. module top_fndcontrol(
3.     input clk,
4.     input [1:0] fndselin,
5.     input [3:0] fndin,
6.     output [3:0] fndsel,
7.     output [7:0] fnd
8. );
9.     mux4x1 U0 (
10.         .s(fndselin),
11.         .y(fndsel)
12.     );
13.     fnddecoder U1 (
14.         .a(fndin),
15.         .fnd(fnd)
16.     );
17. endmodule
    
```

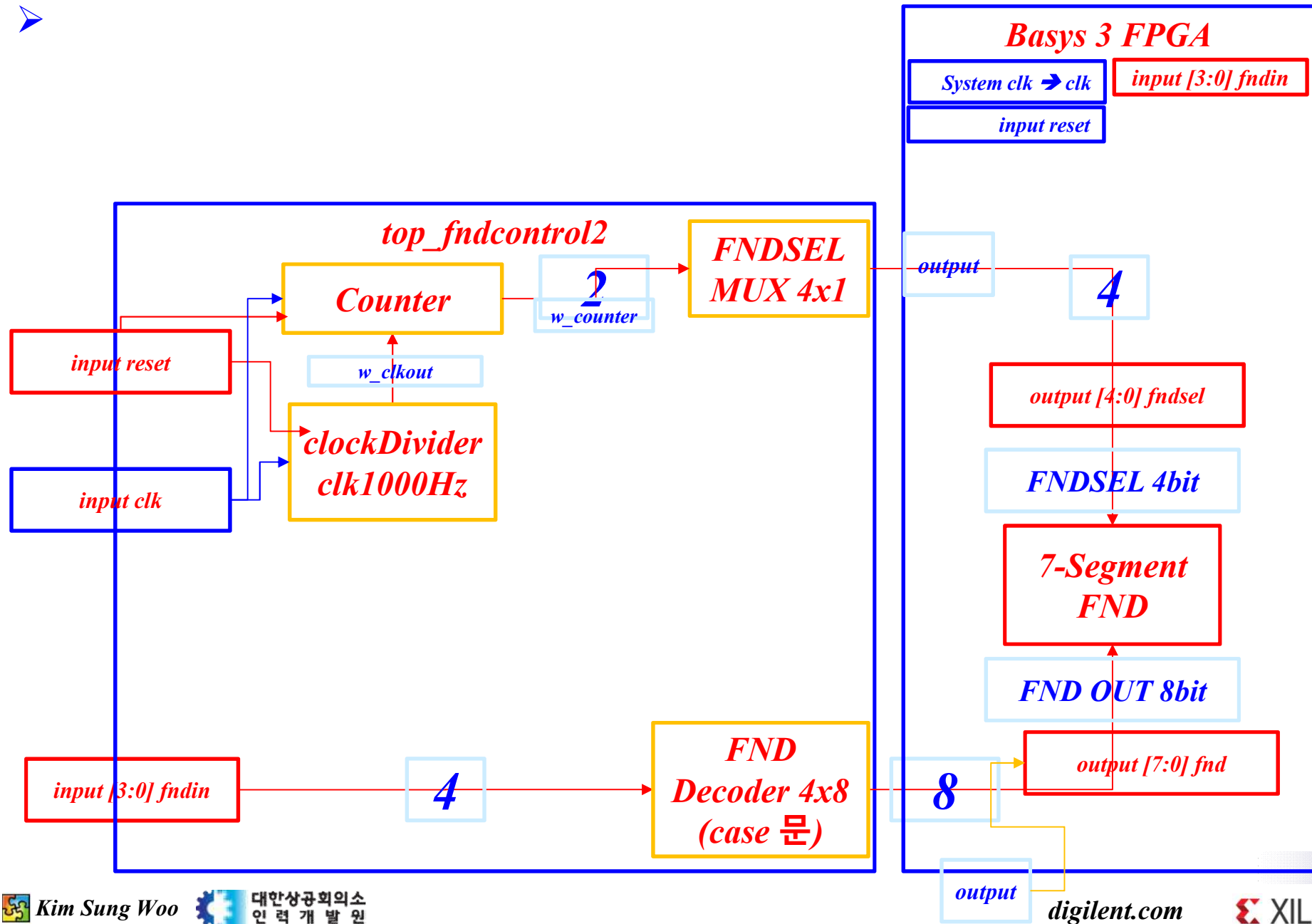


Example 02. FND Controller II



Example 02. *FND Controller II*

Example 02. FND Controller II-1

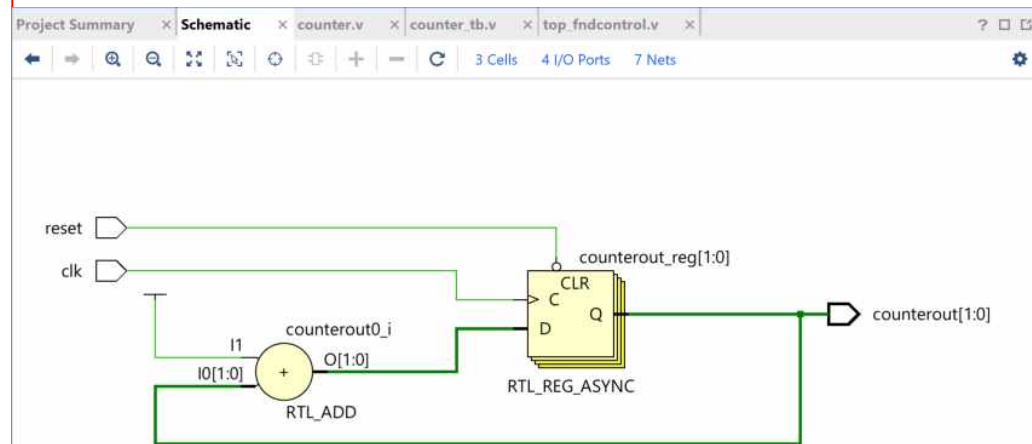


Example 02. FND Controller II-1

➤ counter_exam.xpr

➤ Counter 설계

```
1. `timescale 1ns / 1ps
2. module counter (
3.     input inclk,
4.     input reset,
5.     output reg [1:0] out_counter
6. );
7. always @(posedge inclk, negedge reset) begin
8.     if (~reset) begin
9.         out_counter <= 0;
10.    end else begin
11.        out_counter <= out_counter + 1;
12.    end
13. end
14. endmodule
```



Example 02. FND Controller II-1

➤ counter_exam.xpr

➤ ClockDivider 설계

- 1000Hz Clock

```
1. `timescale 1ns / 1ps
2. module clockdivider (
3.     input clk,
4.     input reset,
5.     output reg clk1000Hz // 1kHz clock
6. );
7. reg [25:0] cnt = 0;
8. always @(posedge clk, negedge reset) begin
9.     if (~reset) begin
10.         cnt <= 0;
11.     end else begin
12.         if (cnt == (50_000 - 1)) begin
13.             cnt <= 0;
14.             clk1000Hz <= ~clk1000Hz;
15.         end else begin
16.             cnt <= cnt + 1;
17.         end
18.     end
19. end
20. endmodule
```

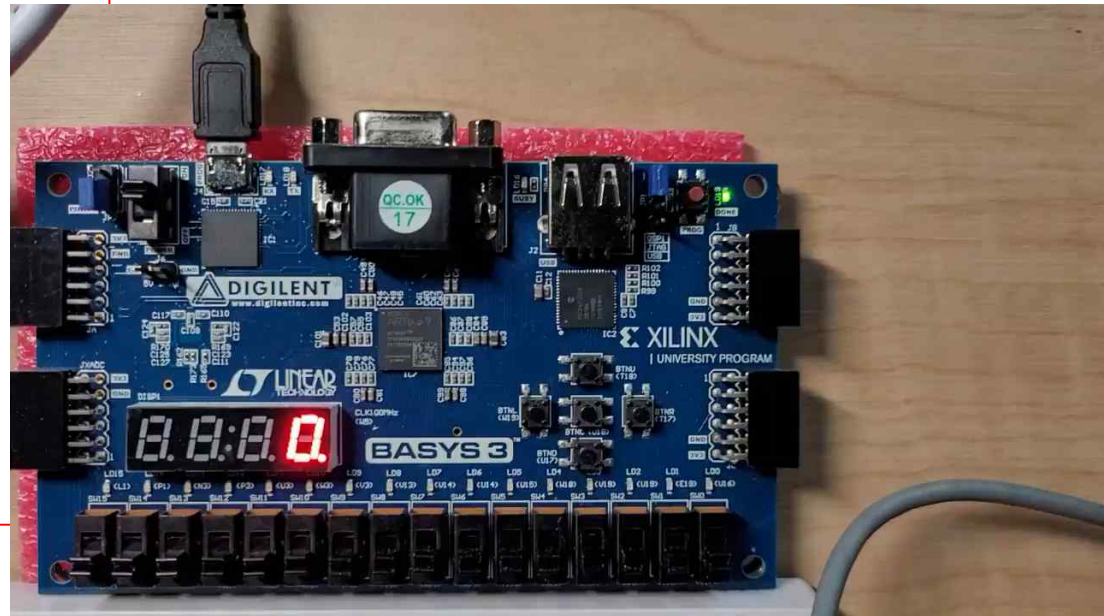
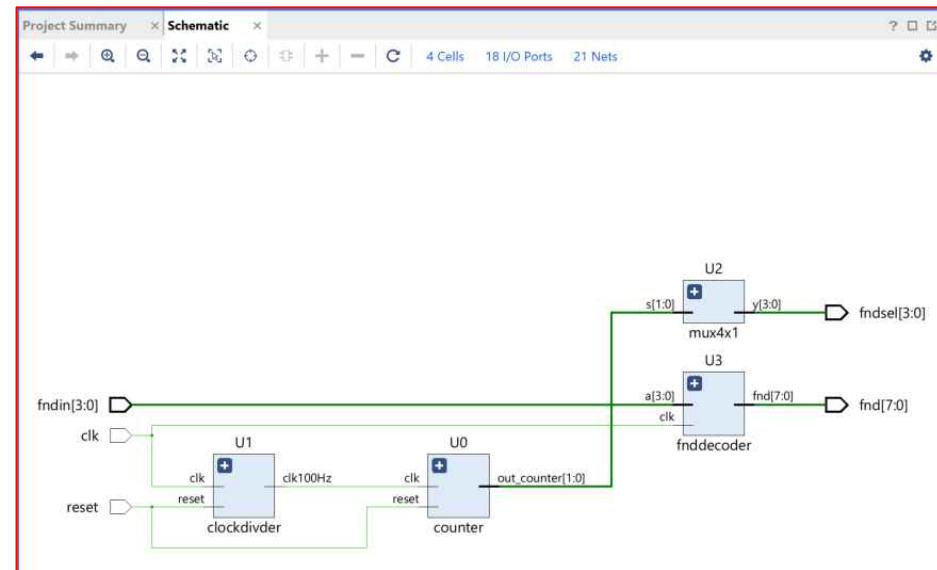
Example 02. FND Controller II-1

➤ counter_exam.xpr

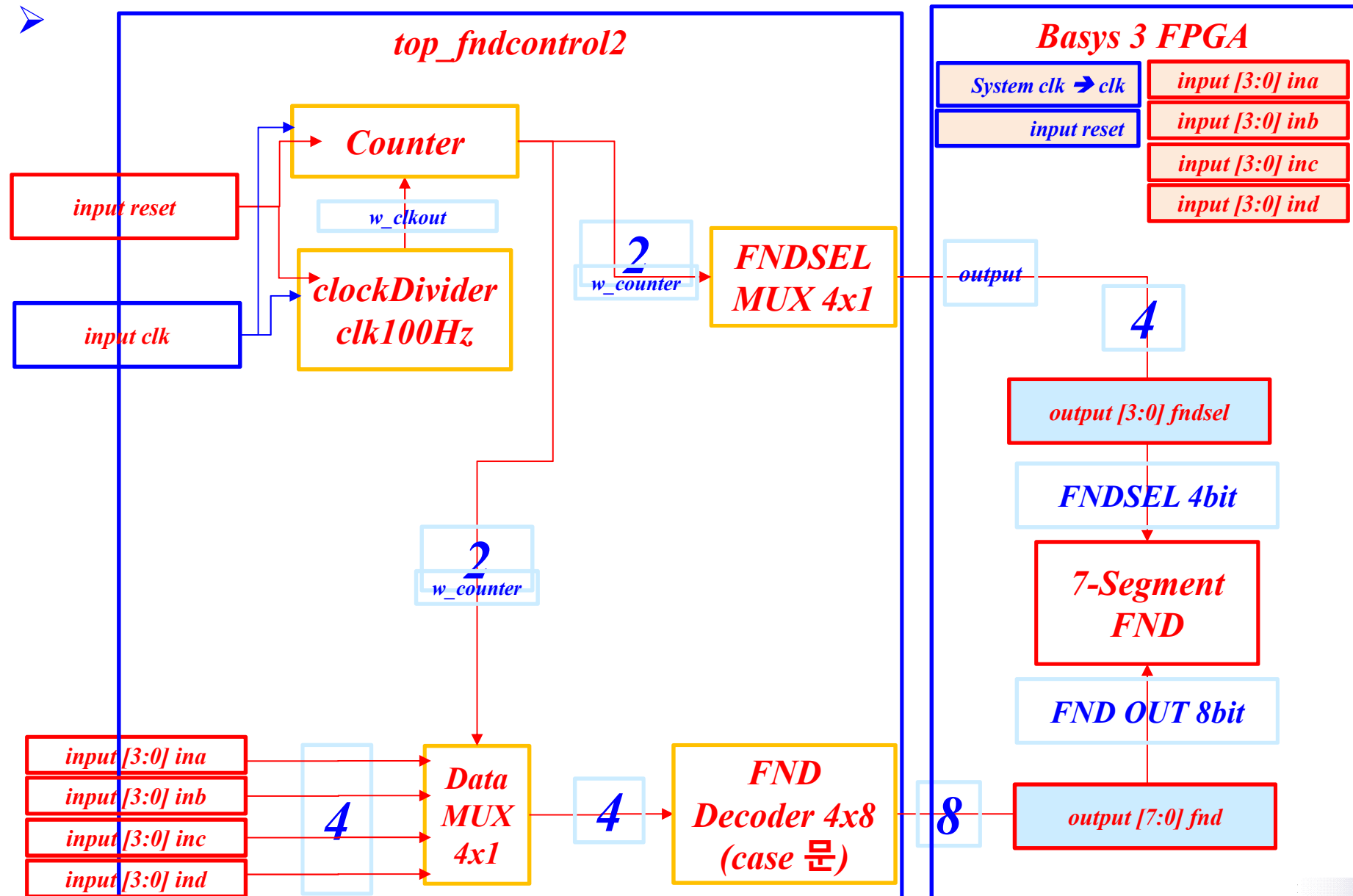


```
1. `timescale 1ns / 1ps
2. module top_fndcontrol2(
3.     input clk,
4.     input reset,
5.     input [3:0] fndin,
6.     output [3:0] fndsel,
7.     output [7:0] fnd
8. );
9. wire [3:0] w_clkout;
10. wire [1:0] w_counter;
11. counter U0 (
12.     .inclk(w_clkout),
13.     .reset(reset),
14.     .out_counter(w_counter)
15. );
```

```
17. clockdivider U1 (
18.     .clk(clk),
19.     .reset(reset),
20.     .clk1000Hz(w_clkout)
21. );
22. mux4x1 U2 (
23.     .s(w_counter),
24.     .y(fndsel)
25. );
26. fnddecoder U3 (
27.     .a(fndin),
28.     .fnd(fnd)
29. );
30. endmodule
```



Example 02. FND Controller II-2

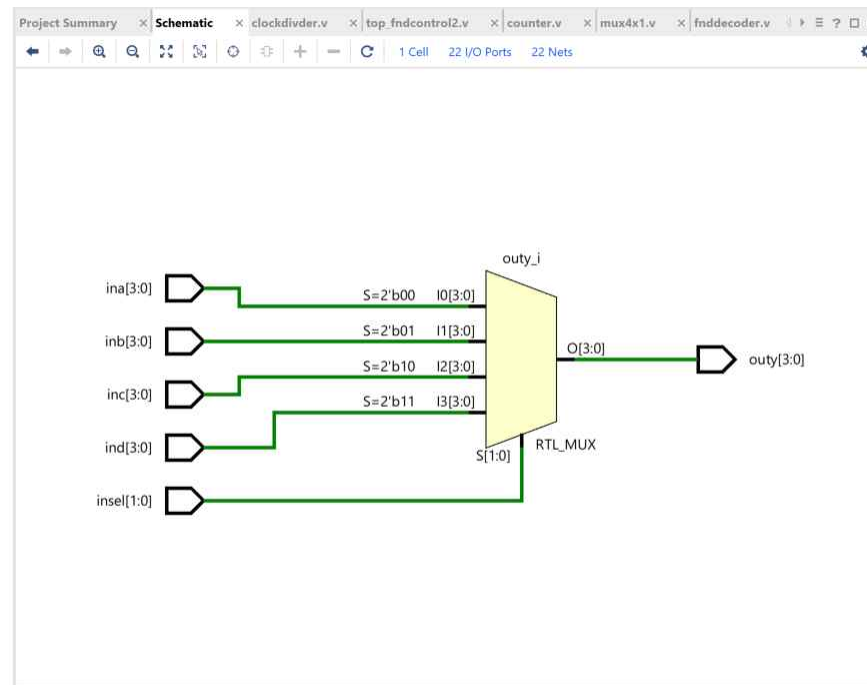


Example 02. FND Controller II

➤ counter_exam.xpr

➤ FND Data 입력 구현을 위하여 **4x1 Data_Mux** 를 이용하여 설계

```
1. `timescale 1ns / 1ps
2. module mux_4x1(
3.     input [3:0] ina, inb,
       inc, ind,
4.     input [1:0] insel,
5.     output reg [3:0] outy
6. );
7.     always @(insel)
8.     begin : datamux4x1
9.         case (insel)
10.            2'b00: outy = ina;
11.            2'b01: outy = inb;
12.            2'b10: outy = inc;
13.            2'b11: outy = ind;
14.        endcase
15.    end
```



Example 02. FND Controller II

➤ counter_exam.xpr

```
1. `timescale 1ns / 1ps
2. module top_fndcontrol2(
3.     input clk,
4.     input reset,
5.     // input [1:0] fndselin,
6.     input [3:0] ina, inb, inc, ind,
7.     output [3:0] fndsel,
8.     output [7:0] fnd
9. );
10. wire w_clkout;
11. wire [1:0] w_counter;
12. wire [3:0] w_datafnd;
13. counter U0 (
14.     .inclk(w_clkout),
15.     .reset(reset),
16.     .out_counter(w_counter)
17. );
```

```
1. clockdivider U1 (
2.     .clk(clk),
3.     .reset(reset),
4.     .clk1000Hz(w_clkout)
5. );
6. datamux4x1 U2 (
7.     .ina (ina),
8.     .inb (inb),
9.     .inc (inc),
10.    .ind (ind),
11.    .insel(w_counter),
12.    .outy (w_datafnd)
13. );
14. mux4x1 U3 (
15.     .s(w_datafnd),
16.     .y(fndsel)
17. );
```

```
1. fnddecoder U4 (
2.     .a(w_datafnd),
3.     .fnd(fnd)
4. );
5. );
6. endmodule
```

